A

Template Sequences

Input Sequence

Increase AA fraction

```
seq_by_classes(90,
aliphatic=0.2)
```

```
seq_by_classes(90,
aliphatic=0.5)
```

Increasing FCR

'change_FCR'

Decreasing net charge

Increasing net charge

'change_NCPR'

Changing charge distribution

'change_kappa'

'change_dimensions'

Shuffle specific residues

'shuffle_specific_residues'

Other chemical properties

B

yeast library

grow to log phase

desiccate on filter

regrowth

plasmid extraction

sequencing

plasmid extraction

C

$\log_2$ (fold change)

Templates

D

Deleterious Protective

-$\log_{10}$ (adj p-val)

$\log_2$ (fold change)

E

Template 6

F

Controls

G

Percentage (%)

n=642

n=334

n=1425

n=6

n=24

n=47

Non-significant

Deleterious

Protective

All

Controls

H

fraction Alanine

Bottom

Top

I

hydrophobicity

J

helix probability

K

ε (self interaction)

L

$\log_2$ (fold change)

complexity

M

fraction Alanine

N

Templates

scaled rg

scaling exponent

fraction M

fraction I

fraction L

complexity

A Patch

Disorder Promoting

fraction A

fraction Aliphatic

Spearman r